Student 18 **Kanigasri M** 192565010RD 5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 3.33 **TC: 2/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    private int rollNo;
    private String name;
    private double marks;
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.rollNo = sc.nextInt();
        sc.nextLine();
        s.name = sc.nextLine();
        s.marks = sc.nextDouble();
        s.displayDetails();
    }
}
```

INPUT

```
012
sam
92.9
```

OUTPUT

```
Roll No: 12
Name: sam
Marks: 92.9
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    private int rollNo;
    private String name;
    private double marks;
    public void setRollNo(int rollNo){
        this.rollNo = rollNo;
    }
    public void setName(String name){
        this.name = name;
    }
    public void setMarks(double marks){
        this.marks = marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
012
sam
89.0
```

OUTPUT

```
Roll No: 12
Name: sam
Marks: 89.0
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo = 101;
        name = "Rahul";
        marks = 85.5;
    }
    Student(int rollNo,String name,double marks){
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s1 = new Student();
        s1.displayDetails();
        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        Student s2 = new Student(rollNo,name,marks);
        s2.displayDetails();
        sc.close();
    }
}
```

INPUT

```
102
sam
89.7
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: sam
Marks: 89.7
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    double basicSalary;
    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
public static void main(String[] args){
    Scanner sc =  new Scanner(System.in);
    Salary s = new Salary();
    int id = sc.nextInt();
    sc.nextLine();
    String name = sc.nextLine();
    double salary = sc.nextDouble();
    s.getEmployeeDetails(id,name,salary);
    s.calculateSalary();
    s.displayDetails();
    sc.close();
}
}
class Salary extends Employee{
    double hra,da,netSalary;
    void calculateSalary(){
        hra = basicSalary*0.20;
        da = basicSalary*0.10;
        netSalary = basicSalary + hra + da;
    }
    void displayDetails(){
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Net Salary: "+netSalary);
    }
}
```

INPUT

```
012
sam
80000
```

OUTPUT

```
Employee ID: 12
Employee Name: sam
Basic Salary: 80000.0
HRA: 16000.0
DA: 8000.0
Net Salary: 104000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    void getEmployeeDetails(int id,String name){
        empId=id;
        empName=name;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        GrossSalary g = new GrossSalary();
        g.getEmployeeDetails(sc.nextInt(),sc.next());
        g.getBasicSalary(sc.nextDouble());
        g.displayDetails();
    }
}
class Salary extends Employee{
    double basicSalary;
    void getBasicSalary(double s){
        basicSalary=s;
    }
}
class GrossSalary extends Salary{
    void displayDetails(){
        double hra=basicSalary*0.2;
        double da=basicSalary*0.1;
        double gross=basicSalary+hra+da;
        System.out.println("Employee ID: "+empId);
        System.out.println("Employee Name: "+empName);
        System.out.println("Basic Salary: "+basicSalary);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+gross);
    }
}
```

INPUT

```
012
sam
80000
```

OUTPUT

```
Employee ID: 12
Employee Name: sam
Basic Salary: 80000.0
HRA: 16000.0
DA: 8000.0
Gross Salary: 104000.0
```